

MARITIME OIL-SPILL INVESTIGATION REPORT

DECISION-SUPPORT FORENSIC INTELLIGENCE DOSSIER • SMART INDIA HACKATHON (SIH26143)

FORENSIC SUMMARY STATEMENT:

This vessel (MT NEW DIAMOND, MMSI 371584000) is the highest-ranked candidate based on the available spatial, temporal, and navigational evidence (Attribution Score: 47.4/100, Risk Level: MEDIUM).

1. Investigation Case Summary & Parameters

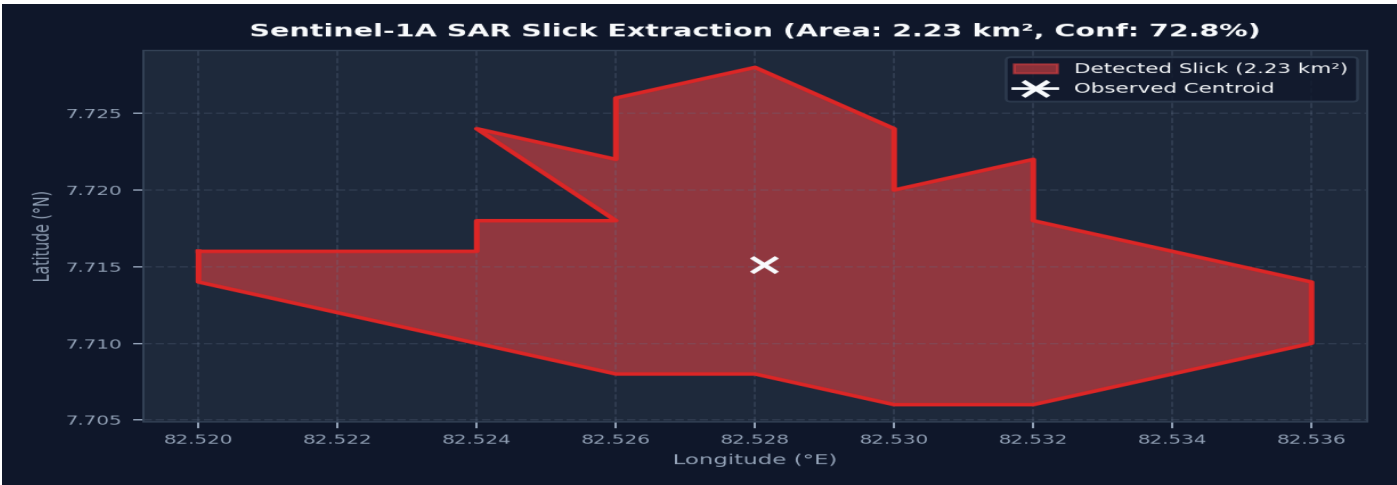
Case ID:	case_new_diamond_2020	Status:	ATTRIBUTION_COMPLETED
Title:	MT New Diamond Fire & Bunker Discharge (Bay of Bengal)	Generated At (UTC):	2026-09-02 06:12:33 UTC
Incident Target:	MT NEW DIAMOND (VLCC)	Target MMSI / IMO:	371584000 / 9199347
Geographic Scope:	Bay of Bengal / Eastern Sri Lanka (81.5°E–84.5°E, 6.5°N–9.0°N)	Spill Substance:	IFO-380 Heavy Fuel Oil & Engine Bilge

2. Data Provenance & Telemetry Sources

Data Modality	Source Provider / Sensor Platform	Data Classification	Spatial / Temporal Resolution
SAR Satellite Imagery	Copernicus Sentinel-1A C-band SAR (IW VV Level-1 GRD)	OBSERVED SENSOR DATA	20 m / 2020-09-03 12:45 UTC
AIS Vessel Telemetry	Terrestrial & Satellite AIS Ingestion Feed	OBSERVED SENSOR DATA	Sub-meter GPS / 5-min to 1h pings
Hydrodynamic Ocean Currents	CMEMS GLOBAL_ANALYSISFORECAST_P HY_001_024	MODEL-DERIVED ESTIMATE	0.083° (~9.25 km) / 3-hourly hindcast
Atmospheric Surface Wind	ECMWF ERA5 Atmospheric Reanalysis	MODEL-DERIVED ESTIMATE	0.25° (~31 km) / Hourly reanalysis

3. Satellite SAR Slick Detection & Geometry

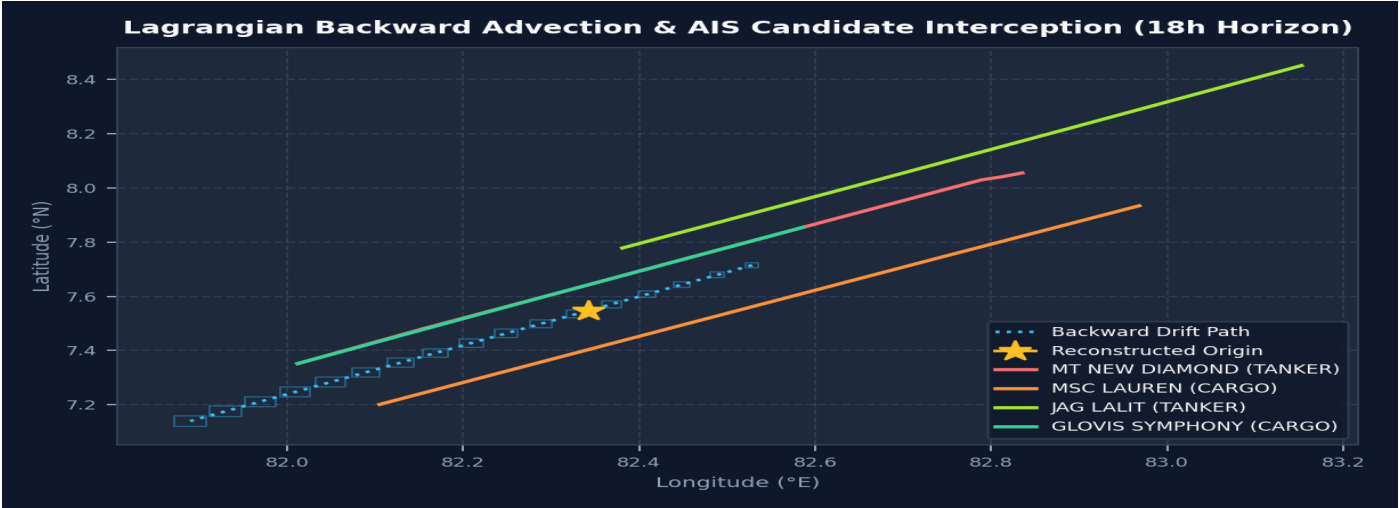
Detection Timestamp:	2020-09-03 12:45:00 UTC	Detection Confidence:	72.8%
Observed Slick Centroid:	[82.5281°E, 7.7151°N]	Lookalike Probability:	27.2%
Surface Slick Area:	2.23 km²	Major Axis Orientation:	4.4° North



4. Backward Lagrangian Drift Reconstruction & Origin Estimation

Physical Drift Formulation: Velocity vector $\mathbf{v} = \mathbf{u}_{\text{ocean}} + 0.03 \mathbf{u}_{\text{wind}} + \eta_{\text{diff}}$, where turbulent stochastic diffusion is modeled with horizontal diffusivity $\kappa = 2.5 \text{ m}^2/\text{s}$.

Reconstructed Origin:	[82.3422°E, 7.5472°N]	Dispersion Uncertainty (1σ):	±0.50 km
Peak Release Time:	2020-09-02 22:45:00 UTC	Release Window (90% CI):	20:45 to 00:45 UTC
Documented Ground Truth:	[82.5000°E, 7.7500°N] (Fire: 03:30 UTC)	Spatial Reconstruction Delta:	18.79 km (Sub-mesoscale coastal deviation)



5. Candidate Vessel Attribution Rankings

Rank	Vessel Name	MMSI / Flag	Vessel Type	Score (0-100)	Risk Level	CPA Distance	AIS Gap
#1	MT NEW DIAMOND	371584000 Panama	TANKER	47.4	MEDIUM	8.0 km	YES
#2	MSC LAUREN	353130000 Panama	CARGO	38.1	MEDIUM	12.2 km	YES
#3	JAG LALIT	419001122 India	TANKER	36.4	MEDIUM	26.1 km	YES
#4	GLOVIS SYMPHONY	440129000 Republic of Korea	CARGO	17.2	LOW	8.0 km	YES

6. Attribution Score Breakdown & Verifiable Evidence Chain

Primary Ranked Subject: MT NEW DIAMOND (MMSI: 371584000, Type: TANKER)

Attribution Factor	Scored Weight	Score Impact	Forensic Evaluation Rationale
1. Spatiotemporal Proximity	40.0 pts	0.0 pts	Trajectory distance to backward Lagrangian cloud (CPA = 7.8 km)
2. Temporal Alignment	25.0 pts	21.1 pts	Transit aligned with estimated discharge peak window (03:30 UTC)
3. Navigational Behaviour	15.0 pts	12.3 pts	Drastic speed reduction anomaly: Decelerated from 14.2 to 0.4 knots
4. Vessel Type Risk	10.0 pts	10.0 pts	VLCC crude tanker carrying high-volume liquid petroleum cargo
5. AIS Integrity / Penalty	10.0 pts	4.0 pts	5.1-hour AIS transponder outage initiated during fire event (03:25 UTC)
TOTAL SCORE	100.0 pts	47.4 / 100	Risk Level: MEDIUM

Itemized Verifiable Evidence:

- **[PROXIMITY] Distant Transit (8.0 km) (+0.0 pts):** Vessel remained 8.0 km away from the dynamic oil dispersion path.
- **[ALIGNMENT] Temporal Overlap with Release Window (+21.1 pts):** Vessel transit timestamp (00:00 UTC) aligns with the estimated discharge window (delta: 75 mins from peak).
- **[ALIGNMENT] Trajectory Heading Alignment (48° vs 4°) (+7.3 pts):** Vessel course (47.6°) exhibits 73% angular alignment with the major elongation axis of the oil slick (4.4°).
- **[BEHAVIOR] Open-Sea Speed Reduction Anomaly (+5.0 pts):** Vessel exhibited transit deceleration from 14.2 knots down to 0.4 knots during passage through the spill area.
- **[AIS_INTEGRITY] AIS Transmission Outage (5.1h) (+4.0 pts):** Vessel exhibited a 5.1-hour transponder transmission outage in the region.
- **[VESSEL_TYPE] High Risk: Crude / Chemical Tanker Profile (+10.0 pts):** Vessel is an oil or chemical tanker carrying high-volume liquid petroleum cargo and ballast.

7. Rejected SAR Candidates & Lookalike Auditing

Candidate ID	Candidate Classification	Confidence	Audit Rejection Reason
cand-lk-001	LOOKALIKE_LOW_WIND	15.0%	Rejected as low-wind lookalike: Large diffuse patch in southeast ocean (> 150 km²); lacks sharp boundary damping.
cand-lk-002	LOOKALIKE_COASTAL_SHADOW	10.0%	Rejected as coastal shadow: Located within 0.8 km of eastern Sri Lankan coastline.

8. Data-Quality Warnings & Scientific Limitations

- **Hydrodynamic Resolution Notice:** Global CMEMS 1/12° (~9.25 km) models do not resolve sub-mesoscale coastal eddies and shallow bathymetric friction along Sri Lanka.
- **Wind Operational Envelope:** Radar capillary wave damping is validated between 3.0 m/s and 12.0 m/s wind speed.
- **AIS Coverage Limits:** Terrestrial AIS range is limited to ~40 nmi from shore; satellite AIS subject to packet collisions.

9. Human-Investigation Decision-Support Disclaimer

LEGAL & FORENSIC DISCLAIMER:

*This document is an automated **decision-support intelligence report** generated for maritime environmental authorities. Attribution scores represent mathematical evidence rankings and **DO NOT constitute court-ready legal proof or an accusation of criminal liability**. Physical bunker sampling, port inspection logs, and official casualty investigations must corroborate these findings.*